Physical Chemistry (I)

Lecture 10

Entropy

Entropy

$$\mathrm{d}S = \frac{\delta q_{\mathrm{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\mathrm{state}\ i}^{\mathrm{state}\ f} \frac{\delta q_{\mathrm{rev}}}{T}$$

- Molar entropy change $\Delta S_{\mathrm{m}} = \Delta S / n$

Examples

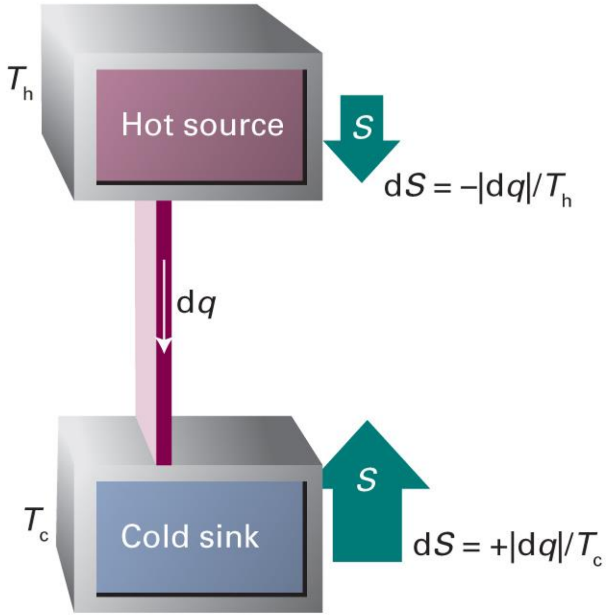
Heat transfer

Expansion (reversible vs. free)

Phase transitions

Heating

Heat Transfer



Transfer of heat $|\delta q|$: h → c

$$dS_h = -|\delta q|/T_h$$

$$dS_c = +|\delta q|/T_c$$

The total entropy change is

$$dS = dS_h + dS_c = |\delta q|\left(\frac{1}{T_c} - \frac{1}{T_h}\right)$$

- Equilibrium: $dS = 0$ or $T_h = T_c$
- Spontaneity: $dS > 0$ or $T_h > T_c$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.13)

Heat Transfer to Surroundings

Consider an infinitesimal transfer of heat

$$\delta q = -\delta q_{\mathrm{sur}}$$

The surroundings are extremely large, so

$$\delta q_{\mathrm{sur}} = \mathrm{d}U_{\mathrm{sur}} \text{ (constant volume)}$$

$$\delta q_{\mathrm{sur}} = \mathrm{d}H_{\mathrm{sur}} \text{ (constant pressure)}$$

Since $\mathrm{d}H = \mathrm{d}U + \mathrm{d}(pV)$, this implies $\mathrm{d}(pV)_{\mathrm{sur}} \approx 0$

Entropy of Surroundings

U and H are state functions, so δq_{sur} is independent of reversibility:

$$\mathrm{d}S_{\mathrm{sur}} = \frac{\delta q_{\mathrm{sur,rev}}}{T_{\mathrm{sur}}} = \frac{\delta q_{\mathrm{sur}}}{T_{\mathrm{sur}}}$$

As the temperature is invariant,

$$\Delta S_{\mathrm{sur}} = \frac{q_{\mathrm{sur}}}{T_{\mathrm{sur}}}$$

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Entropy Calculation

- Heat transfer to surroundings can be always considered as reversible heat transfer

$$\delta q_{\mathrm{sur}} = \delta q_{\mathrm{sur,rev}} \text{ (always)}$$

- Reversible change:

$$\mathrm{d}S = \delta q_{\mathrm{rev}}/T = \delta q/T$$

$$\mathrm{d}S_{\mathrm{sur}} = \delta q_{\mathrm{sur,rev}}/T = -\delta q/T$$

- Irreversible change:

$$\mathrm{d}S \neq \delta q/T \text{ and } \mathrm{d}S_{\mathrm{sur}} = -\delta q/T$$

Reversible Expansion

$$\delta q = -\delta q_{\mathrm{sur}}$$

$$\Delta S = \frac{q_{\mathrm{rev}}}{T}$$

$$\Delta S_{\mathrm{sur}} = \frac{q_{\mathrm{sur}}}{T} = -\frac{q_{\mathrm{rev}}}{T} = -\Delta S$$

Hence,

$$\Delta S_{\mathrm{total}} = \Delta S + \Delta S_{\mathrm{sur}} = 0$$

Free Expansion

For a perfect gas, isothermal expansion accompanies with

$$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right)$$

However, $w = 0$ (free expansion) and $\Delta U = 0$ (perfect gas)

$$q = q_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = 0$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} > 0$$

Reversible Phase Transition

Normal transition temperature T_{trs}: the temperature at which two phases are in equilibrium at 1 atm

If the system is at T_{trs} and 1 atm,

$$q = \Delta_{\text{trs}}H \text{ (constant pressure)}$$

$$\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{sur}} = -\Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{total}} = \Delta_{\text{trs}}S + \Delta_{\text{trs}}S_{\text{sur}} = 0$$

Heating at Constant V

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant volume, $\delta q_{\text{rev}} = \mathrm{d}U = C_V \mathrm{d}T$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_V}{T} \mathrm{d}T$$

For constant C_V,

$$S(T_f) = S(T_i) + C_V \int_{T_i}^{T_f} \frac{1}{T} \mathrm{d}T = S(T_i) + C_V \ln\left(\frac{T_f}{T_i}\right)$$

Heating at Constant p

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant pressure, $\delta q_{\text{rev}} = \mathrm{d}H = C_p \mathrm{d}T$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_p}{T} \mathrm{d}T$$

For constant C_p,

$$S(T_f) = S(T_i) + C_p \int_{T_i}^{T_f} \frac{1}{T} \mathrm{d}T = S(T_i) + C_p \ln\left(\frac{T_f}{T_i}\right)$$

Entropy as State Function

$$
\begin{tikzcd}
\text{A state} \arrow[r, "\Delta S_{\mathrm{AB}}"] \arrow[d, "\Delta S_{\mathrm{AC}}"'] & \text{B state} \\
\text{C state} \arrow[r, "\Delta S_{\mathrm{CD}}"'] & \text{D state} \arrow[u, "\Delta S_{\mathrm{DB}}"']
\end{tikzcd}
$$

$$\Delta S_{\mathrm{AB}} = \Delta S_{\mathrm{AC}} + \Delta S_{\mathrm{CD}} + \Delta S_{\mathrm{DB}}$$

Example 3B.1

Calculate the entropy change when argon at 25 °C and 1.00 bar in a container of volume 0.500 dm^3 is allowed to expand to 1.000 dm^3 and is simultaneously heated to 100 °C. (Take the molar heat capacity at constant volume to be $(3/2)R$.)

Stage 1: isothermal expansion at 25 °C, 0.500 dm^3 → 1.000 dm^3

$$\Delta S(\text{stage 1}) = nR\ln\left(V_f/V_i\right)$$

Stage 2: isochoric heating at 1.000 dm^3, 25 °C → 100 °C

$$\Delta S(\text{stage 2}) = nC_{V,\mathrm{m}}\ln\left(T_f/T_i\right) = (3/2)nR\ln\left(T_f/T_i\right)$$

Example 3B.1

$n = pV/RT$

$= (1.00\times10^5 \text{ Pa}) \times (0.500\times10^{-3} \text{ m}^3) / (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) / 298 \text{ K}$

$= 0.0201 \text{ mol}$

$\Delta S(\text{stage 1}) = nR \ln(V_f/V_i)$

$= 0.0201 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times \ln(1.000/0.500)$

$= 0.116 \text{ J K}^{-1}$

$\Delta S(\text{stage 2}) = (3/2)nR \ln(T_f/T_i)$

$= (3/2) \times 0.0201 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times \ln(373/298)$

$= 0.056 \text{ J K}^{-1}$

Example 3B.1

Therefore,

$$\Delta S = \Delta S(\text{stage 1}) + \Delta S(\text{stage 2})$$

$$= 0.116 \text{ J K}^{-1} + 0.056 \text{ J K}^{-1} = 0.173 \text{ J K}^{-1}$$

Nernst Heat Theorem

"The entropy change accompanying any transformation approaches zero as the temperature approaches zero:

$$\Delta S \rightarrow 0 \text{ as } T \rightarrow 0,$$

if all the substances involved are perfectly ordered."

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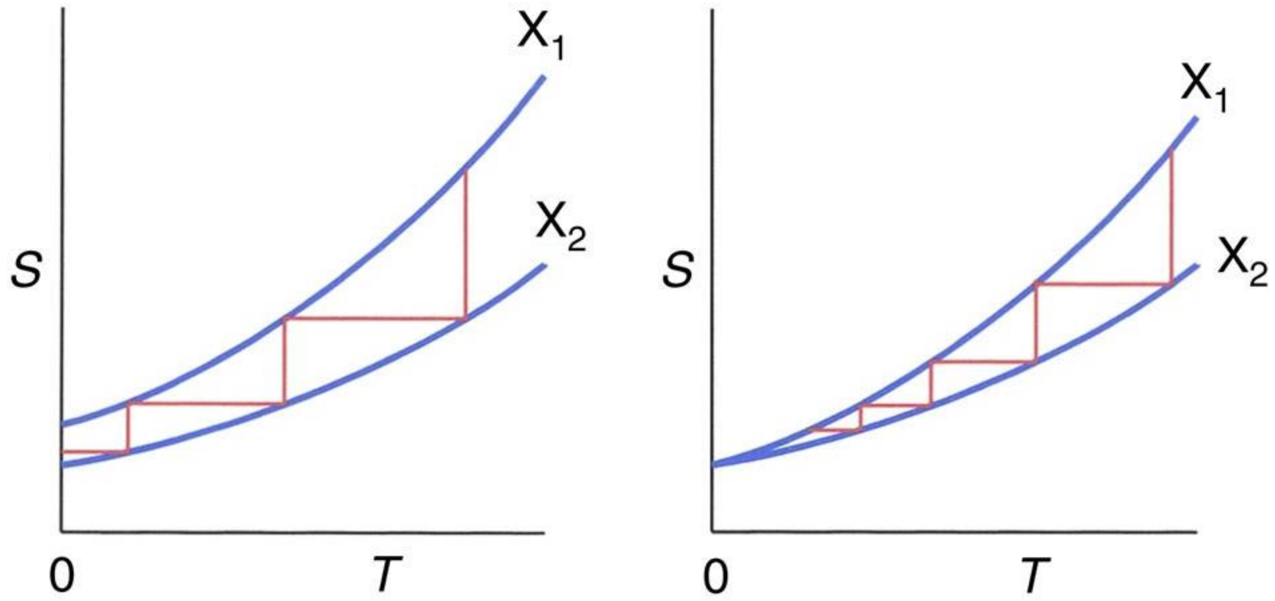
Unattainability Principle



(Image: Masanes and Oppenheim, *Nature Communications 8*: 14538 (2017), Figure 1)

Brief Illustration 3C.2

Calculate the entropy of the transition between orthorhombic sulfur, α, and monoclinic sulfur, β, at 0 K.

$$\begin{tikzcd}[column sep=huge, row sep=huge]
\boxed{\alpha \text{ (at 0 K)}} \arrow[r, "\Delta_{\mathrm{trs}}S(0)"] \arrow[d, "S_{\alpha}(T_{\mathrm{trs}}) - S_{\alpha}(0)"'] & \boxed{\beta \text{ (at 0 K)}} \\
\boxed{\alpha \text{ (at } T_{\mathrm{trs}})} \arrow[r, "\Delta_{\mathrm{trs}}S(T_{\mathrm{trs}})"'] & \boxed{\beta \text{ (at } T_{\mathrm{trs}})} \arrow[u, "S_{\beta}(0) - S_{\beta}(T_{\mathrm{trs}})"']
\end{tikzcd}$$

Brief Illustration 3C.2

1. α (at 0 K) ⟶ α (at T_{trs})

$$S_{\alpha,\mathrm{m}}(T_{\mathrm{trs}}) - S_{\alpha,\mathrm{m}}(0) = \int_{0}^{T_{\mathrm{trs}}} \frac{C_{p,\mathrm{m}}}{T}\,\mathrm{d}T = 37\ \mathrm{J\ K^{-1}\ mol^{-1}}$$

2. α (at T_{trs}) ⟶ β (at T_{trs})

$$\Delta_{\mathrm{trs}}S_{\mathrm{m}}(T_{\mathrm{trs}}) = \frac{\Delta_{\mathrm{trs}}H_{\mathrm{m}}(T_{\mathrm{trs}})}{T_{\mathrm{trs}}} = \frac{402\ \mathrm{J\ mol^{-1}}}{369\ \mathrm{K}} = 1.09\ \mathrm{J\ K^{-1}\ mol^{-1}}$$

3. β (at T_{trs}) ⟶ β (at 0 K)

$$S_{\beta,\mathrm{m}}(0) - S_{\beta,\mathrm{m}}(T_{\mathrm{trs}}) = \int_{T_{\mathrm{trs}}}^{0} \frac{C_{p,\mathrm{m}}}{T}\,\mathrm{d}T = -38\ \mathrm{J\ K^{-1}\ mol^{-1}}$$

Brief Illustration 3C.2

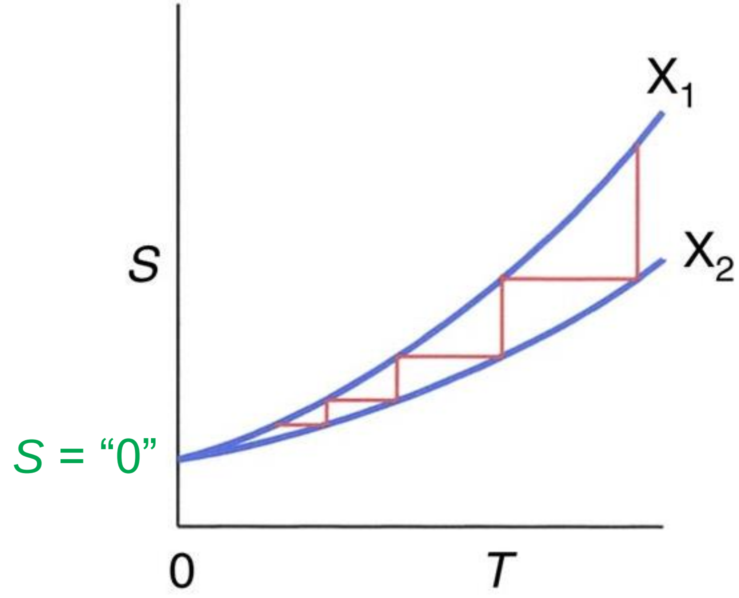
Therefore,

$$\Delta_{\mathrm{trs}}S(0) = \{S_{\alpha,\mathrm{m}}(T_{\mathrm{trs}}) - S_{\alpha,\mathrm{m}}(0)\} + \Delta_{\mathrm{trs}}S_{\mathrm{m}}(T_{\mathrm{trs}}) + \{S_{\beta,\mathrm{m}}(0) - S_{\beta,\mathrm{m}}(T_{\mathrm{trs}})\}$$

$= 37\ \mathrm{J\ K^{-1}\ mol^{-1}} + 1\ \mathrm{J\ K^{-1}\ mol^{-1}} - 38\ \mathrm{J\ K^{-1}\ mol^{-1}} = 0$

Third Law of Thermodynamics

"The entropy of all perfect crystalline substances is zero at T = 0."



(Image: Masanes and Oppenheim, *Nature Communications* 8: 14538 (2017), Figure 1)

Third-Law Entropy

$$S(T) - \underbrace{S(0)}_{=0} = \int_0^T \frac{C_p(T')}{T'}\,\mathrm{d}T'$$

$$S(T) = \int_0^T \frac{C_p(T')}{T'}\,\mathrm{d}T'$$

Debye T^3 Law

For a crystalline solid at $T \sim 0$,

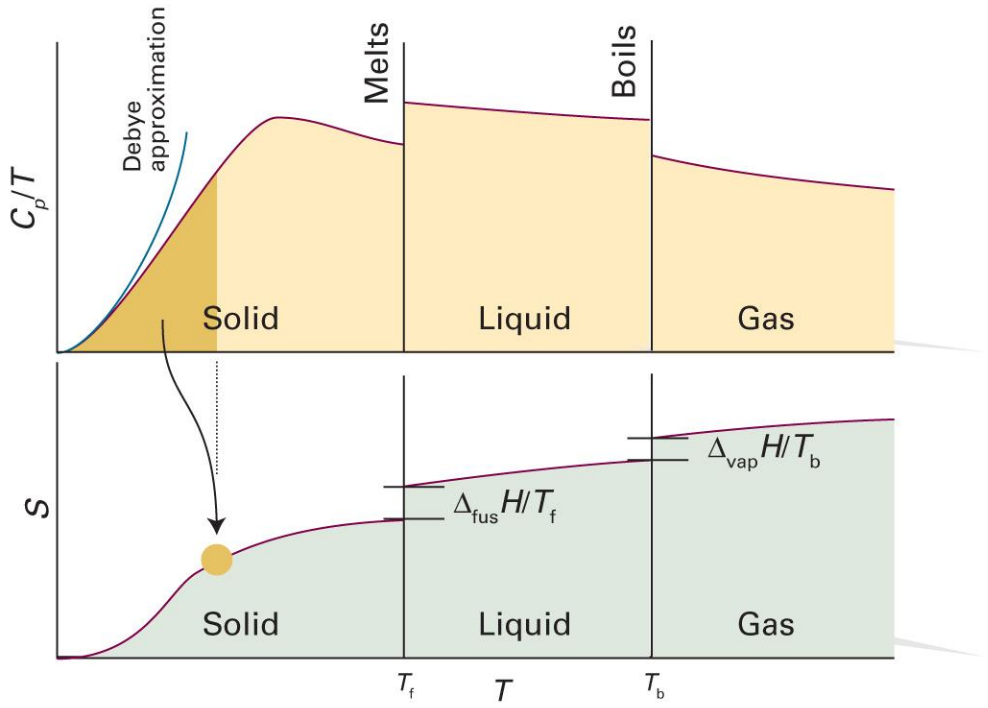
$$C_{p,\mathrm{m}}(T) = aT^3$$

Hence,

$$S_\mathrm{m}(T) = \int_0^T \frac{C_{p,\mathrm{m}}(T')}{T'}\mathrm{d}T' = \int_0^T \frac{a(T')^3}{T'}\mathrm{d}T' = \frac{1}{3}aT^3 = \frac{C_{p,\mathrm{m}}}{3}$$

when T is close to zero.

Typical Entropy Curve



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3C.1)

Thermochemistry of Entropy

- Standard entropy $S^{\ominus}$: third-law entropy of a pure substance at 1 bar (always positive)

- Molar standard entropy $S_{\mathrm{m}}^{\ominus} = S^{\ominus}/n$

- Standard reaction entropy $\Delta_{\mathrm{r}}S^{\ominus}$

$$\Delta_{\mathrm{r}}S^{\ominus} = \sum_{\text{products } i} \nu(i)S_{\mathrm{m}}^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)S_{\mathrm{m}}^{\ominus}(j)$$

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Enthalpy vs. Entropy

- Enthalpy: no natural absolute value

$$\Delta_\mathrm{r}H^\ominus = \sum_{\text{products } i} \nu(i)\Delta_\mathrm{f}H^\ominus(i) - \sum_{\text{reactants } j} \nu(j)\Delta_\mathrm{f}H^\ominus(j)$$

- Entropy: third-law absolute value

$$\Delta_\mathrm{r}S^\ominus = \sum_{\text{products } i} \nu(i)S_\mathrm{m}^\ominus(i) - \sum_{\text{reactants } j} \nu(j)S_\mathrm{m}^\ominus(j)$$

One Exception

$S^{\ominus}(\mathrm{H}^{+}, \mathrm{aq}) = 0$: convention

The entropies of ions in water are values relative to the hydrogen ion in water

→ can either be positive or negative

Temperature Dependence

$$S_{\mathrm{m}}(T_2) = S_{\mathrm{m}}(T_1) + \int_{T_1}^{T_2} \frac{C_{p,\mathrm{m}}(T)}{T}\,\mathrm{d}T$$

Using the definition of standard reaction entropy,

$$\Delta_{\mathrm{r}}S^{\ominus}(T_2) = \Delta_{\mathrm{r}}S^{\ominus}(T_1) + \int_{T_1}^{T_2} \frac{\Delta_{\mathrm{r}}C_p^{\ominus}(T)}{T}\,\mathrm{d}T$$

Similar to Kirchhoff's law!

Temperature Dependence

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \int_{T_1}^{T_2} \frac{\Delta_r C_p^\ominus(T)}{T} \mathrm{d}T$$

If $\Delta_r C_p^\ominus$ is constant,

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \int_{T_1}^{T_2} \frac{1}{T} \mathrm{d}T$$

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \ln(T_2/T_1)$$

So, what does entropy mean?

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Second Law

$\mathrm{d}S \geq 0$ (for an isolated system)

- $\mathrm{d}S = 0$: reversible change, equilibrium
- $\mathrm{d}S > 0$: irreversible change, spontaneous change

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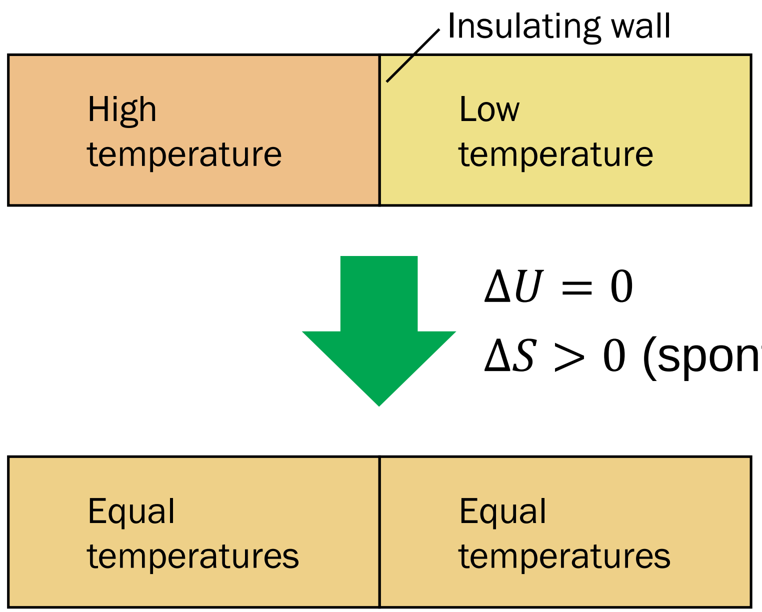
Entropy and Work



$\Delta U = 0$

$\Delta S > 0$ (spontaneous)

Loss of capacity to do work

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Index of Exhaustion

Entropy can be considered as an index of exhaustion

- Clausius: “The energy of the universe is constant; the entropy tends to a maximum.”
- Heat death of the universe: when S_{universe} is maximized
- Jeremy Rifkin, *Entropy: A New World View* (1980)

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Statistical Consideration

- (Macro)state: defined by values of macroscopic variables
- Microstate: specific microscopic arrangement
- Different microstates can be mapped into the same macrostate

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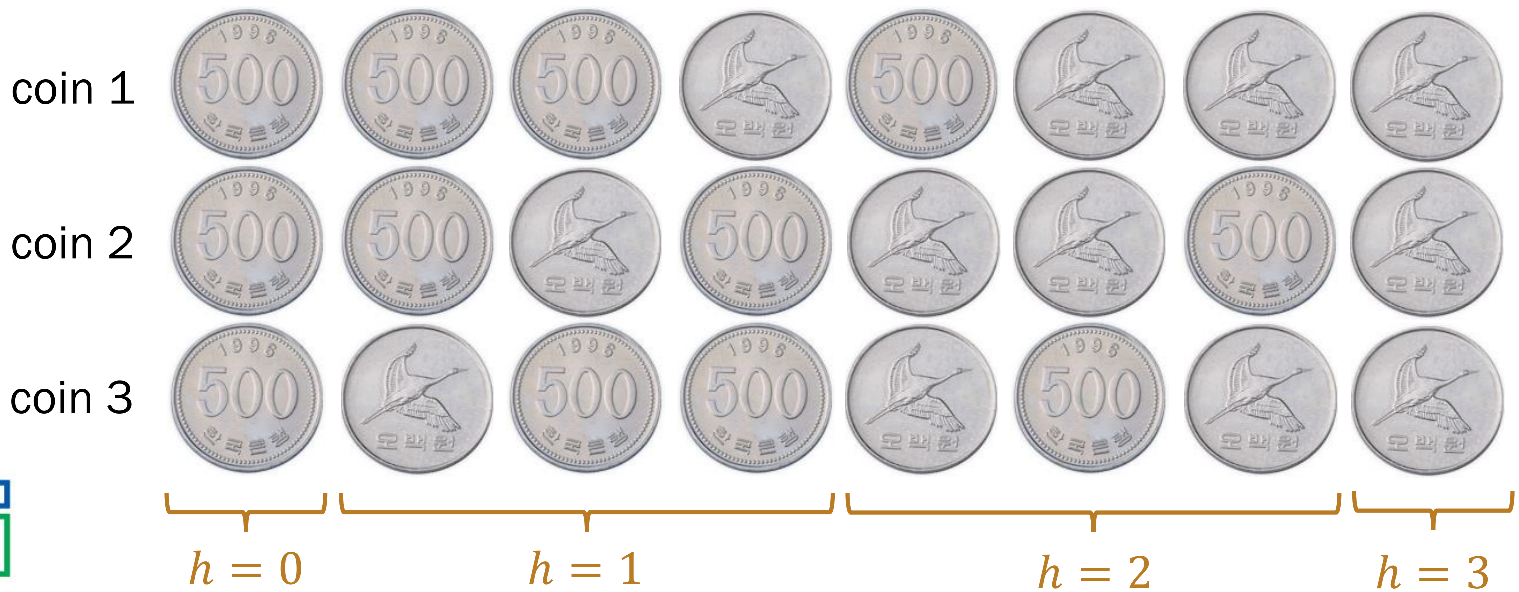
Example: Coin Toss

When tossing three coins,

- Macrostate: defined by the number of heads h
- Microstate: the actual arrangement of coins



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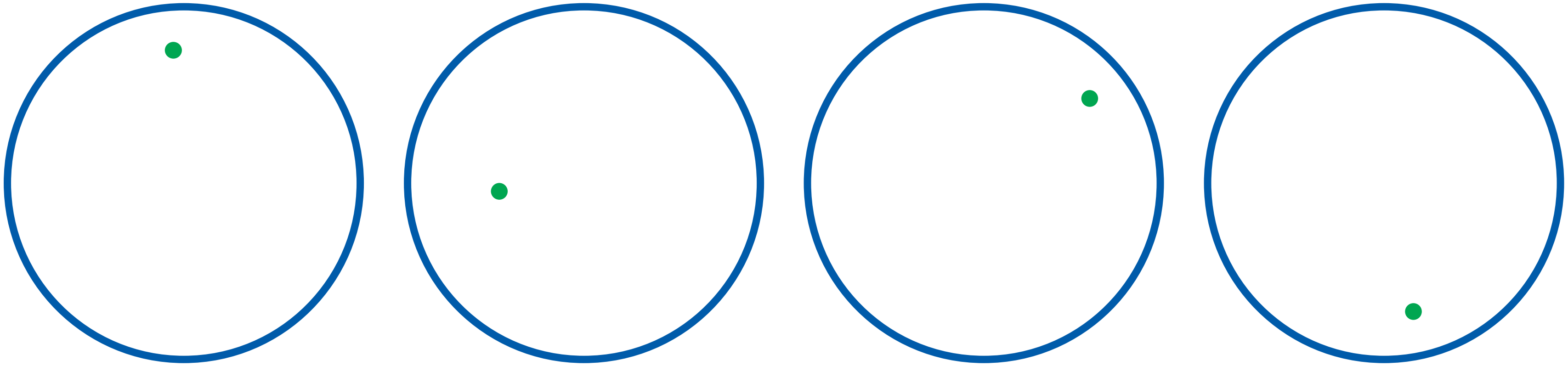
Example: Diffusion

When a gas molecule moves around the system,

- Macrostate: defined by the volume V
- Microstate: the position of the molecule



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Number of Microstates *W*

"The number of microstates

corresponding to a specific macrostate"

- Coin toss

$W(h=0)=1, W(h=1)=3, W(h=2)=3, W(h=3)=1$

- Diffusion

Let the volume occupied by a molecule v_0;

for the system volume V, $W(V)=V/v_0$

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Boltzmann's Formula

$$S = k_{\mathrm{B}} \ln W$$

number of microstates

Boltzmann constant (1.38 × 10^{-23} J/K)

$$\Delta S = S_f - S_i = k_{\mathrm{B}} \ln W_f - k_{\mathrm{B}} \ln W_i = k_{\mathrm{B}} \ln\left(\frac{W_f}{W_i}\right)$$

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Example: Diffusion

Suppose that the system volume changes from V_i to V_f:

$$W_i = V_i / v_0$$

$$W_f = V_f / v_0$$

The change in entropy is given by

$$\Delta S_1 = k_{\mathrm{B}} \ln\left(W_f / W_i\right) = k_{\mathrm{B}} \ln\left(V_f / V_i\right)$$

For a perfect gas consisting of N molecules,

$$\Delta S_N = N k_{\mathrm{B}} \ln\left(V_f / V_i\right) = nR \ln\left(V_f / V_i\right)$$

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Second Law Revisited

$\mathrm{d}S \geq 0$ (for an isolated system)

As $S = k_{\mathrm{B}} \ln W$ is a monotonic function of W,

$\mathrm{d}W \geq 0$ (for an isolated system)

The number of microstates never decreases

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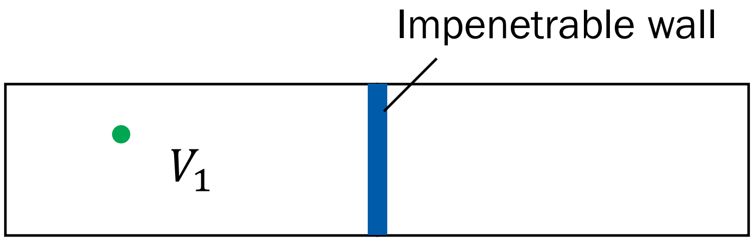
Example: Diffusion

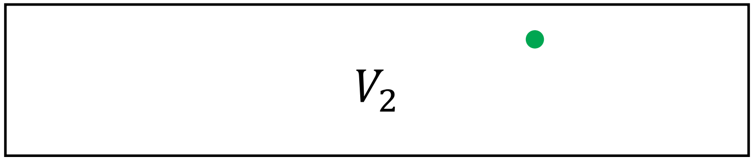


$W_1 = V_1/v_0$

$S_1 = k_\mathrm{B} \ln W_1$

$\Delta W > 0$

$\Delta S > 0$ (spontaneous)



$W_2 = V_2/v_0$

$S_2 = k_\mathrm{B} \ln W_2$

Third Law Revisited

At $T = 0$,

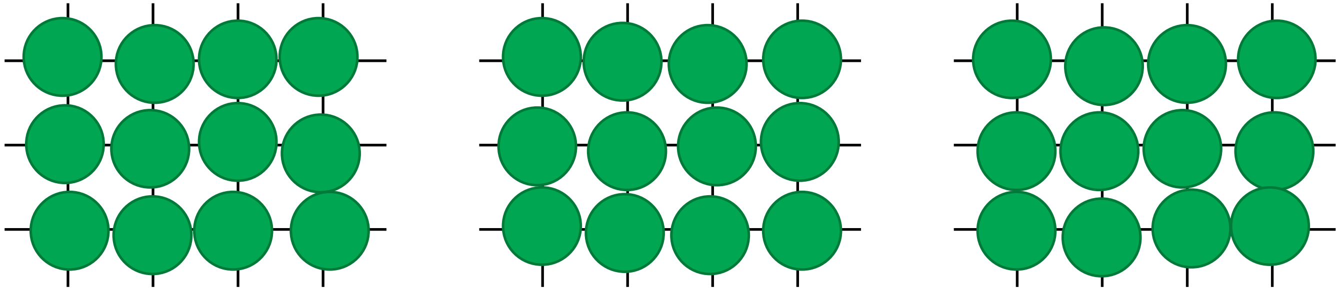
$$S = k_{\mathrm{B}} \ln W = 0$$

$$W = 1$$

Only one microstate is possible at 0 K

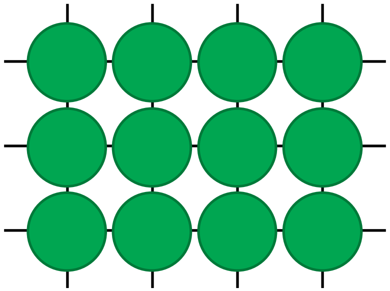
Solid at Absolute Zero

$$T > 0$$



...

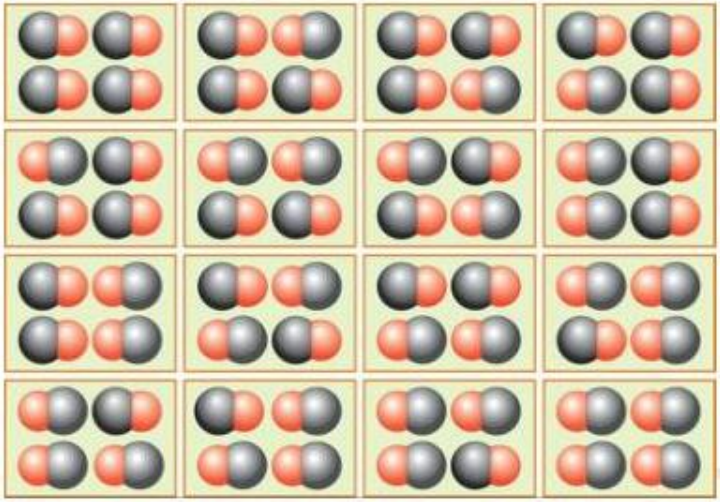
$$T = 0$$



Residual Entropy

$W > 1$ and $S > 0$ even at $T = 0$

e.g. carbon monoxide (CO)



(Image: https://image2.slideserve.com/4094967/slide18-l.jpg)

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Molecular View on Entropy

In general, W is larger (and S is greater) for a molecule with

1. more atoms

$$\mathrm{CO} < \mathrm{CO_2}$$

2. more flexibility

$$\mathrm{CH_3CH_2CH_2CH_2CH_3} > \begin{array}{c} \mathrm{CH_3} \\ | \\ \mathrm{H_3C{-}C{-}CH_3} \\ | \\ \mathrm{CH_3} \end{array}$$

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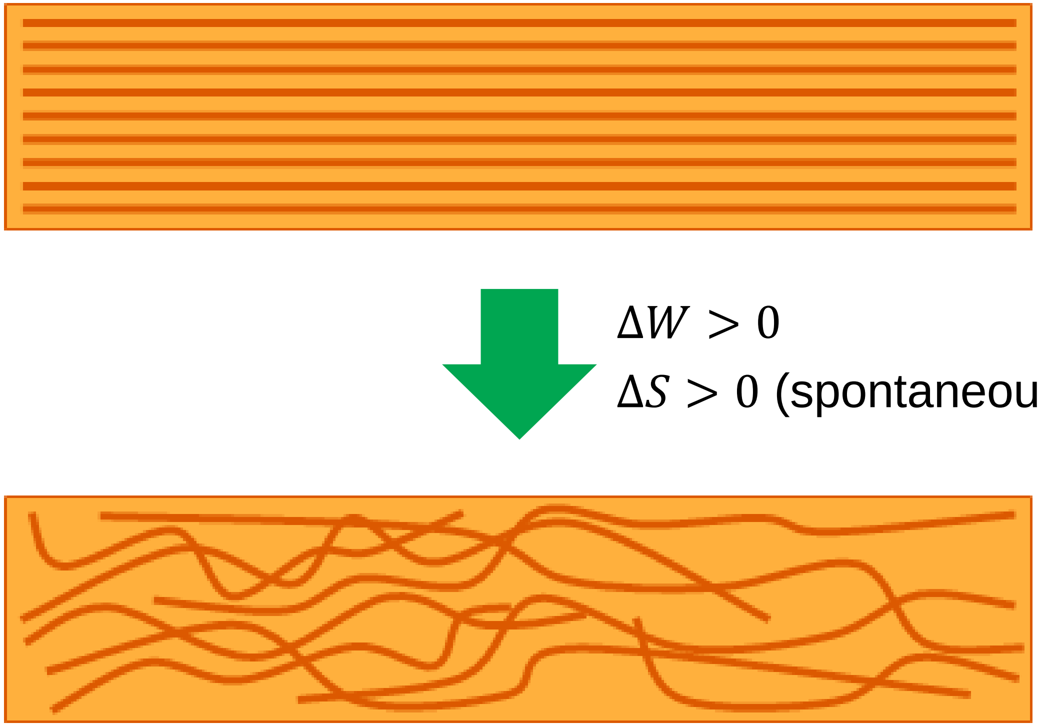
Polymer Conformations



$\Delta W > 0$

$\Delta S > 0$ (spontaneous)

(Image: https://www.schoolphysics.co.uk/age14-16/Matter/text/Rubber_band/index.html)